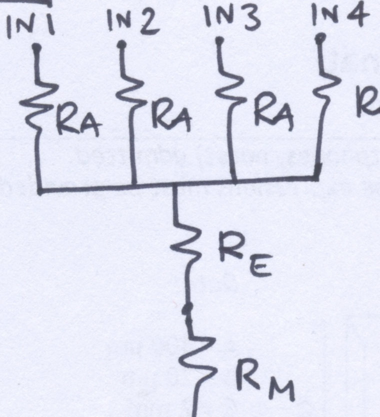
EX. 1

a)



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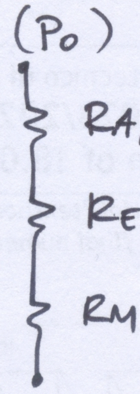
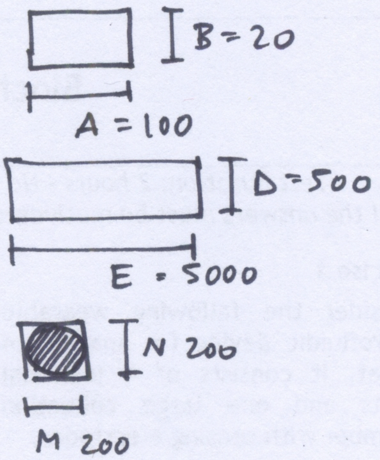


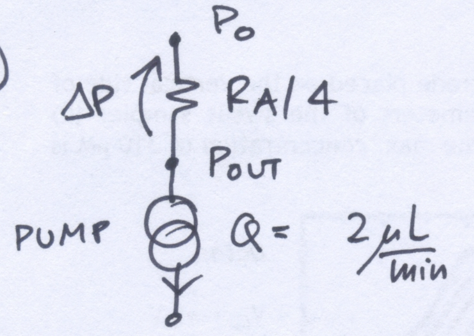


$$R_A = \frac{12 \cdot \eta \cdot \ell^{C}}{\left(1-0.63\frac{B}{A}\right) B^3 A} = \frac{12 \cdot 10^{-3} \cdot 2 \cdot 10^{-3}}{(1-0.63/5)\cdot(20\mu m)^3 \cdot 100\mu m} = \boxed{3.43 \cdot 10^{13} \left[\frac{Pa \cdot s}{m^3}\right]}$$

$$R_E = \frac{12 \cdot \eta \, F}{\left(1-\frac{0.63}{10}\right)(500\mu m)^3 \cdot 5 mm} = \boxed{4 \cdot 10^{7} \left[\frac{Pa \cdot s}{m^3}\right]} \rightarrow \text{NEGLIGIBLE}$$

$$R_M = \frac{8 \cdot \eta \, H}{\pi \left(\frac{M}{2}\right)^4} = \boxed{2.54 \cdot 10^{9} \left[\frac{Pa \cdot s}{m^3}\right]} \rightarrow \text{NEGLIGIBLE}$$

b)



$$P_{OUT} = P_0 - \Delta P = P_0 - \frac{R_A}{4} \cdot Q$$

$$= 101\,kPa - \underbrace{\frac{3.43}{4} \cdot 10^{13} \cdot \frac{2 \cdot 10^{-6} \cdot 10^{-3}}{60} \frac{m^3}{s}}_{285,8\ Pa}$$

$$= 100.7\,kPa$$

$$V_A = \frac{Q}{4} \cdot \frac{1}{B \cdot A} = \boxed{4.2\ mm/s}$$

$$V_E = Q \cdot \frac{1}{D \cdot E} = \boxed{1.3 \cdot 10^{-5}\ m/s}$$

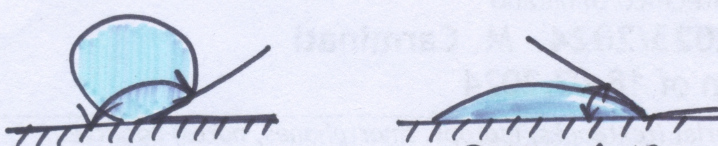
$$V_M = Q \, \frac{1}{M \cdot N} = \boxed{8.3 \cdot 10^{-4}\ m/s}$$

$$\text{fill time} = \frac{\text{Volume}}{Q} = \frac{D \cdot E \cdot F}{Q} = \boxed{150\ s.}$$

c)

	PROS	CONS
DIAPHARM	INFINITE PUMPING VOLUME EASY ON-CHIP INTEGRATION	PULSATILE FLOW
SYRINGE	CONSTANT FLOW	FINITE VOLUME

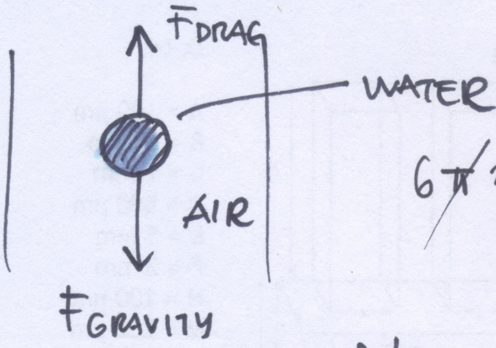
d) BAD WETTABILITY GOOD WETTABILITY



LARGE ANGLE SMALL ANGLE

$\longrightarrow$ FOR CAPILLARY FLOW

e)



GRAVITY ARCHIMEDE'S PUSH

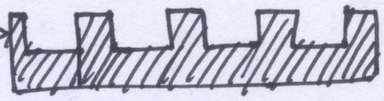
$$6\pi\eta\, \cancel{r}\, V_{terminal} = \underbrace{\left(\rho_{WATER} - \rho_{AIR}\right)}_{} \cdot g \cdot \frac{4}{3}\cancel{\pi} r^{\cancel{3}\,2}$$

$\rho_{AIR} \rightarrow$ negligible $\frac{1}{1000}$

$$V_{terminal} = g \cdot \frac{4}{3} \cdot \frac{1}{6} \cdot \left(\frac{\rho_{WATER} \cancel{\rho_{AIR}}}{\eta}\right) r^2$$

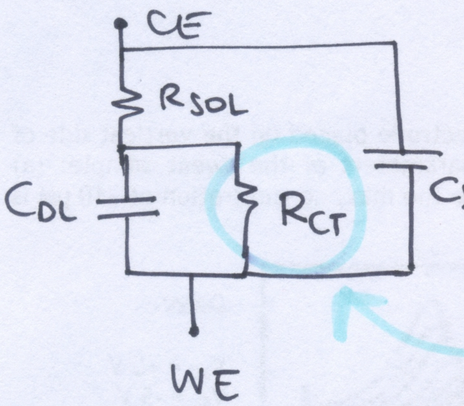
$$= \boxed{3\ mm/s}$$

f) MOULD $\rightarrow$



EX. 2

a)



$$C_{DL} = 0.1 \frac{pF}{\mu m^2} \cdot 600\mu m \cdot 200\mu m = 12\ nF$$

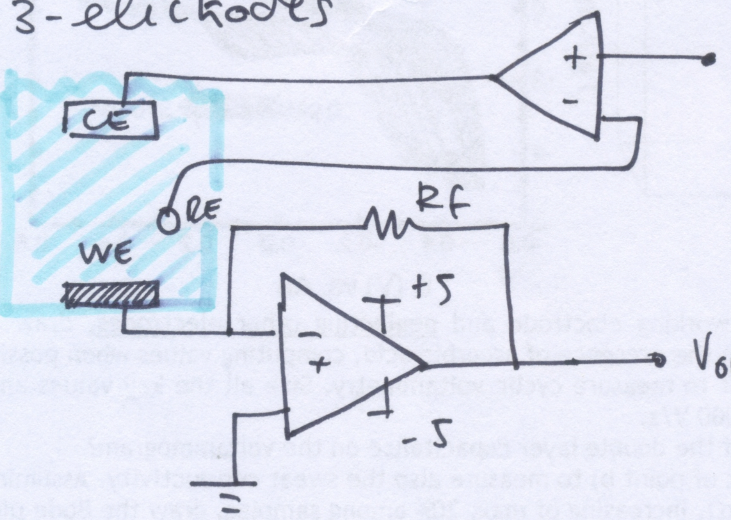
R_{CT} from voltammogram $R_{CT} = f(E)$

R_{SOL} = need geometry

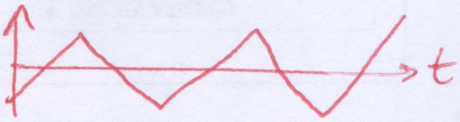
C_{SOL} = " "

REDOX

b) CV $\rightarrow$ 3-electrodes



$V_{IN} \in [\pm 0.6V]$



from VOLTAMMOGRAM : $I_{redox} = \pm 4 \frac{\mu A}{mm^2} \cdot 0.6mm \cdot 0.2mm = 480\ nA$

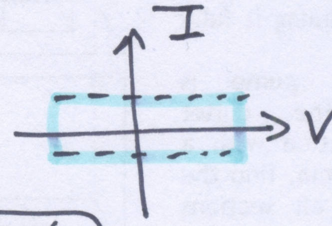
MAX CONCENTRATION

$$\begin{cases} \text{SCAN RATE} \rightarrow \text{BW} \\ \text{SATURATION} \rightarrow V_{OUT}\big|_{max} \end{cases} \quad \begin{cases} BW = \dfrac{1}{2\pi R_f \cdot 0.2pF} = 40 \cdot SR = 40\,kHz \\ 5V = R_f\, I_{max} \end{cases}$$

$$\begin{cases} \dfrac{1}{2\pi R_f \cdot 0.2pF} \geq 40\,kHz \\ R_f \cdot 480nA < 5V \end{cases} \quad \begin{cases} R_f < \dfrac{1}{2\pi\, 40kHz \cdot 0.4pF} \simeq 19M\Omega \\ R_f < \dfrac{5V}{480nA} = 10\,M\Omega \end{cases}$$

MOST STRINGENT: $\boxed{R_f = 10\,M\Omega}$

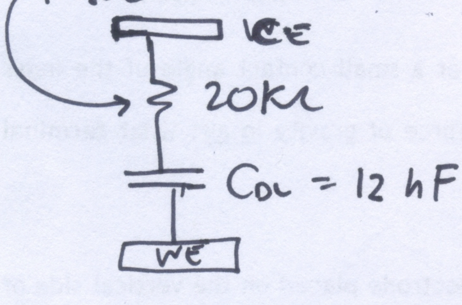
c) $i = C \cdot \dfrac{dV}{dt} = C_{DL} \cdot SR \longrightarrow$

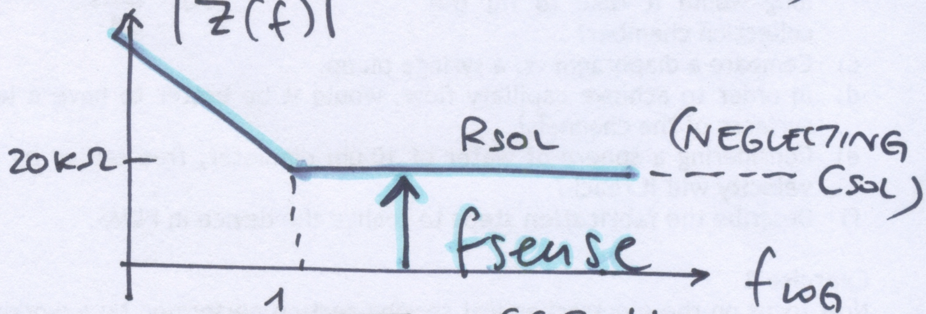


OFFSET CURRENT CHARGING
(C_{DL} at fixed current)

IMPEDANCE → 2 electrodes $\quad V_{IN} = V_{AC}$

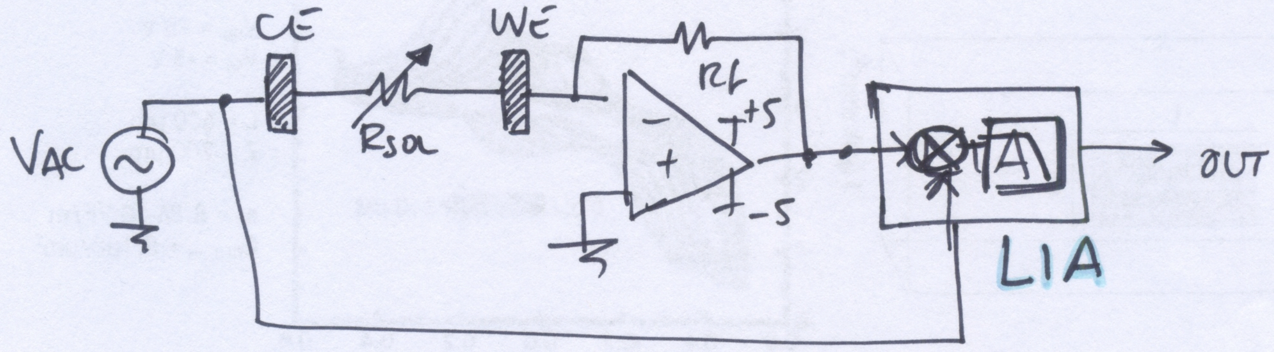
d) $R_{SOL} = 20k\Omega$





$$\frac{1}{2\pi C_{DL} R_{SOL}} = 663\ Hz$$

$f_{sense} > 663Hz \Longrightarrow f_{sense} = 6\,kHz < 40\,kHz$ compatible



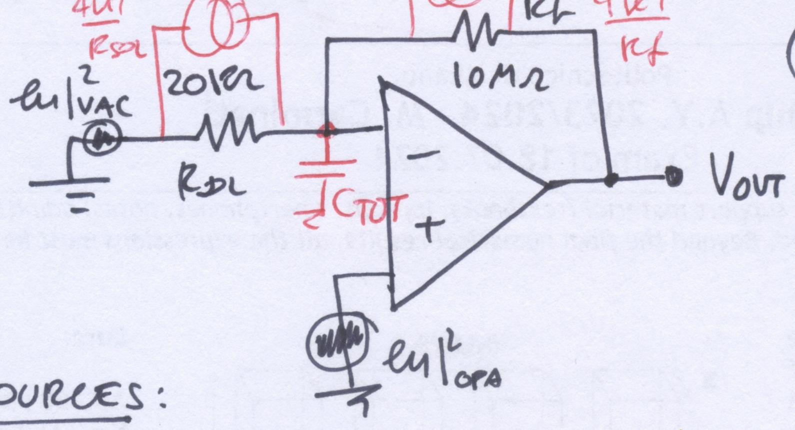
if I keep $R_f = 10M\Omega \longrightarrow$ what V_{AC}?

$|V_{OUT}| = V_{AC} \cdot \dfrac{R_f}{R_{SOL}} \qquad R_{SOL}\big|_{min} = 20k\Omega$

$5 = V_{AC} \cdot \dfrac{10M\Omega}{20k\Omega} \Rightarrow V_{AC} = \dfrac{5 \cdot 20\,k\Omega}{10\,M\Omega} = 10\,mV$ ACCEPTABLE

(with the same R_f of CV, we sense impedance at 6kHz)

e) (neglect C_{DL} at high frequency) 4/4



NOISE SOURCES:

- VAC $\longrightarrow$ $e_n^2|_{VAC}$ = NOT GIVEN
- OPAMP $\longrightarrow$ $e_n^2|_{OPA} = (2\,nV/\sqrt{Hz})^2$ — C_{TOT} NOT GIVEN, I assume $C_{TOT} = 10pF$
- RESISTORS THERMAL NOISE — R_{SOL}, R_f

$$S_i\Big|_{TOT} = \cancel{\frac{4kT}{R_f}} + \frac{4kT}{R_{SOL}} + \cancel{\left(\frac{e_n|_{OPA}}{R_f}\right)^2} + \left(\frac{e_n|_{OPA}}{R_{SOL}}\right)^2 + \left(\frac{e_n|_{VAC}}{R_{SOL}}\right)^2$$

$$+ \left(2\pi f\, C_{TOT} \cdot e_n|_{OPA}\right)^2 = \ldots$$

- $\frac{4kT}{R_{SOL}} = 4\frac{kT}{q}\cdot q\,\frac{1}{R_{SOL}} = \left(0.91\frac{pA}{\sqrt{Hz}}\right)^2$

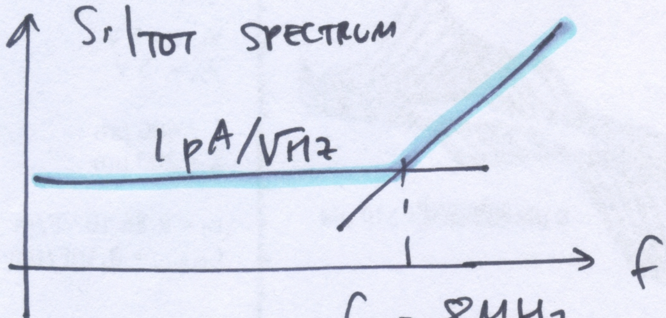
- $\left(\frac{e_n|_{OPA}}{R_{SOL}}\right)^2 = \left(0.1\frac{pA}{\sqrt{Hz}}\right)^2$

if $C_{TOT} = 10pF$

f_c =

$$2\pi f_c \cdot 10pF \cdot \frac{2nV}{\sqrt{Hz}} = .1\frac{pA}{\sqrt{Hz}}$$

$$f_c = 8\,MHz$$



f) RESISTIVE SENSING OR CAPACITIVE: